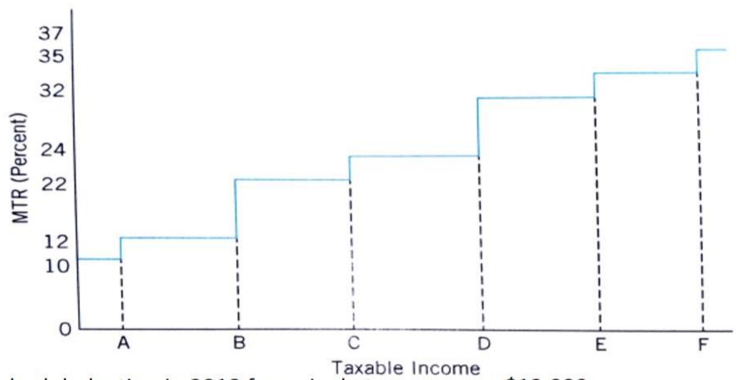
財政學 HW5

1. (Hyman 14.3) A single worker has gross income subject to tax of $40,000. She makes a $5,000 contribution to a special tax-deferred retirement plan offered by her employer. The worker claims one personal exemption for herself and has the following deductible payments: $1,000 in mortgage interest, $1,000 in state income tax, and $500 in property tax. Does it pay the worker to itemize deductions when filing her 2018 tax return? Using the 2018 tax rate schedule shown in Figure 14.1, calculate the worker's tax liability

FIGURE 14.1 Statutory MTRs for the U.S. Personal Income Tax, 2018*



The standard deduction in 2018 for a single taxpayer was $12,000
The U.S. personal income tax has a seven-bracket progressive rate structure.
A = $9,525 for single taxpayers, $19,050 for married taxpayers filing jointly.
B = $38,700 for single taxpayers, $77,400 for married taxpayers filing jointly.
C = $82,500 for single taxpayers, $165,000 for married taxpayers filing jointly.
D = $157,500 for single taxpayers, $315,000 for married taxpayers filing jointly.
E = $200,000 for single taxpayers, $400,000 for married taxpayers filing jointly.
F = $500,000 for single taxpayers, $600,000 for married taxpayers filing jointly.
*Intervals on the horizontal axis are not drawn to scale.

ANS

The standard deduction in 2018 for a single taxpayer was $12,000. It therefore does not pay for this taxpayer to itemize deductions because her itemized deductions amount to only $2,500. The worker's adjusted gross income is $35,000. Her taxable income is $23,000, which is obtained by subtracting the $12,000 standard deduction from the adjusted gross income. The tax is
.10($9,525) + 0.15($13,475) = $952.5 + $2,021.25= $2,973.75 (Rounded to $2,974.00)

2. (Hyman 14.5) Suppose the current progressive income tax structure is scrapped and replaced by a 15 percent tax on all income with no exemptions or deductions allowed except that the first $10,000 of income will not be subject to taxation. Explain why this so-called flat-rate tax is really still a progressive-rate structure. How will the elimination of tax preferences affect resource allocation and prices in markets? How

could the reform, if it really simplified the complexity of the tax code, save resources? What are some common objections to the flat-rate tax?

ANS

The tax rate structure is a two-step progressive rate with a zero rate applying to the first bracket up to $10,000 and 15 percent applying to income above that amount. Elimination of tax preferences will reduce the excess burden of the income tax, but distortions in labor and capital markets will remain. The prices of tax-preferred activities are likely to be affected. For example, home prices could fall because of the elimination of the tax deductibility of mortgage interest. Interest rates paid for borrowing by state and local government would rise because that interest income would no longer be tax exempt. Similarly, there will be many shifts in demand and supply affecting the market prices of a slew of previously tax-preferred activities. To the extent to which the tax code is simplified, compliance costs will be reduced. Fees paid to tax preparers, accountants, and tax lawyers will fall and be reallocated for use elsewhere. Typical objections to the flat-rate tax are based on notions of vertical equity. Many believe that the upper-income groups should pay higher percentages of their income in taxes than lower-income groups do. A shift from the current tax structure to the flat-rate tax will clearly increase the after-tax income of the upper-income groups and increase the tax burden on the lower-income groups.

3. (Rosen 17.3) Singh, who has a federal personal income tax rate of 28 percent, holds an oil stock that appreciates in value by 10 percent each year. He bought the stock one year ago. Singh's stockbroker now wants him to switch the oil stock for a gold stock that is equally risky. Singh has decided that if he hold the oil stock, he will keep it only one more year and then sell it. If he sells the oil stock now, he will invest all the (after-tax) proceeds of the sale in the gold stock and then sell the gold stock one year from now. What is the minimum rate of return the gold stock must pay for Singh to make the switch? Relate your answer to the lock-in effect.

ANS

Suppose Singh buys the oil stock for $1,000 at the start of period 0. At the start of period 1, he has two options: A) hold the oil stock one more period, then sell or B) sell the oil stock, buy the gold stock, and hold it for one period. In both cases, it is assumed that all assets are sold, and any taxes paid at the end of period.

What are the returns to Option A? At a 10 percent rate of appreciation, the oil stock is worth $1,210 after period 2, the capital gain is $210 and assuming a 28 percent rate applies to capital gains, the capital gains tax is 28 percent of \$210 or \$58.80. Thus, Singh is left with \$1,210 - \$58.80 or \$1,151.20 after tax.

If Singh follows Option B, the value of the oil stock at the start of period 1 is \$1,000, the capital gain is \$100, and the tax \$28. Thus Singh has \$1072 left over to proceeds (V) from selling the gold stock at the end of period 1.

V = Value of gold stock – taxes

= (1+r)(\$1,072) –(.28)[(1+r)(\$1,072)-(\$1,072)]

= (1+r)(\$1,072) –(.28)r(\$1,072)

= \$1,072 + .72r(\$1,072)

Setting V = \$1,151.20 (same as oil stock):

\$1,151.20 = \$1,072 + .72r(\$1.072) ⇒ r=10.26%

Thus the gold stock must pay a "premium" because the unrealized capital gains in the oil stock are treated preferentially by the tax code. This is the "lock-in" effect.

4. (Rosen 17.6) Suppose that a typical taxpayer has a marginal income tax rate of 35 percent. The nominal interest rate is 13 percent, and the expected inflation rate is 8 percent.

a. What is the real after-tax rate of interest?

b. Suppose that the expected inflation rate increases by 3 percentage points to 11 percent, and the nominal interest rate increases by the same amount. What happens to the real after-tax rate of return?

c. If the inflation rate increases as in part b, by how much would the nominal interest rate have to increase to keep the real after-tax interest rate at the same level as in part a? Can you generalize your answer using an algebraic formula?

ANS

a. With a tax rate of $t=0.35$, and a nominal interest rate of $i=0.13$, the nominal after-tax rate of interest is $(1-t)i=(1-0.35)0.13=0.0845$. With an expected inflation rate of $\pi=0.08$, the real after-tax rate of interest is $(1-t)i-\pi=(1-0.35)0.13-0.08=0.0845-0.08=0.0045$.

b. If the expected inflation rate increased by 3 percentage points to $\pi=0.11$, and the nominal interest rate also increases by 3 percentage points to $i=0.16$, then the real after-tax rate of interest is now $(1-t)i-\pi=(1-0.35)0.16-0.11=0.104-0.11=-0.006$. The real after-tax rate of return falls from 0.45% to -0.6%. This is because the tax system taxes *nominal*, not real, returns. A person actually loses money in real terms.

c. In general, consider two rates of inflation, $\pi<\pi'$. The key question is when taxes are present, by how much must the nominal interest rate increase in order to have the same real rate of return. This is calculated by equating real rates of return under different inflation rates (holding constant taxes): $(1-t)i-\pi=(1-t)i'-\pi'$. One can rearrange this equation: $(\pi'-\pi)=(1-t)(i'-i)$, or $(i'-i)= (\pi'-\pi)/(1-t)$. This can in turn be expressed as: $\Delta i=\Delta\pi/(1-t)$. Intuitively, the left-hand side of this final equation is the change in nominal interest rates that keeps the real after-tax rate of interest unchanged. It is equal to the change in the expected inflation rate divided by (1-t). Thus, returning to part *6b*, for a 3 percentage point change in the inflation rate and a marginal tax rate of $t=0.35$, nominal interest rates would have to increase by approximately 4.6 percentage points.

5. 根據中華民國 114 年度綜合所得稅結算申報書與所得稅法，請回答以下問題：

(1) 目前的標準扣除額額度為多少？
(2) 請列出我國綜合所得稅的列舉扣除額項目
(3) 請列出我國綜合所得稅的特別扣除額項目

ANS

(1) 單身者 131,000 元，與配偶合併申報者 262,000 元。
(2) 捐贈、保險費、醫藥及生育費、災害損失、購屋借款利息
(3) 財產交易損失、薪資所得特別扣除、儲蓄投資特別扣除、身心障礙特別扣除、教育學費特別扣除、幼兒學前特別扣除、長期照顧特別扣除、房屋租金支出特別扣除

6. 請列出 114 年綜合所得稅最高級距累進差額的計算過程

ANS

590000*(40-5)%+(1330000-590000)*(40-12)%+(2660000-1330000)*(40-20)%+(4980000-2660000)*(40-30%)=911700